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BIOLOGY

Biology: Life Processes — Part 3

Class 12

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Chapter 1

The Living World

What distinguishes the living from the non-living? Living organisms exhibit growth, reproduction, metabolism, response to stimuli and cellular organization. The diversity of life is organized through taxonomy into kingdoms, divisions, classes, orders, families, genera and species. Binomial nomenclature, introduced by Carolus Linnaeus, gives each organism a two-part scientific name. Museums, herbaria, botanical gardens and zoological parks preserve and study biodiversity. Understanding the living world is the foundation of biology and essential for medicine, agriculture and conservation.

Chapter 2

Cell: The Unit of Life

The cell is the basic structural and functional unit of life. The cell theory, proposed by Schleiden and Schwann, states that all living things are composed of cells and that the cell is the basic unit of life. Cells are classified as prokaryotic (without a nucleus, e.g., bacteria) or eukaryotic (with a true nucleus). The cell membrane, cytoplasm and organelles such as the nucleus, mitochondria, endoplasmic reticulum, Golgi apparatus and ribosomes each perform specialized functions. Plant cells differ from animal cells in having cell walls, plastids and large vacuoles. The discovery of the cell revolutionized our understanding of biology.

Chapter 3

Human Physiology

Human physiology studies how the body functions. Digestion breaks down food into absorbable nutrients; respiration releases energy from these nutrients. The circulatory system transports oxygen, nutrients and waste using the heart and blood vessels. The excretory system removes nitrogenous wastes through the kidneys. The nervous and endocrine systems coordinate and regulate body activities. Homeostasis maintains internal stability. Understanding physiology is the basis of medicine and health science.